

COINDESK RESEARCH NOTE

Bitfarms: A race against time

Matt Yamamoto

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Industry wide, bitcoin miners are facing pressure from increased network difficulty and reduced block reward subsidies following the halving. Publicly traded Canadian mining company Bitfarms is no different. At face value, one might look at the financials of Bitfarms and see a miner with fast-growing operations at positive operating cash flows. Extend that viewpoint out a year and a half, however, and that assertion might turn out differently as the company's balloon payments on its high-interest rate debt come due in 2021.

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COMPANY PROFILE

As the bitcoin price was reaching its peak, Bitfarms began its cryptocurrency mining operations in November 2017. Located in Quebec, an area known for its cool temperatures and plentiful amounts of hydroelectricity, Bitfarms manages five facilities constructed from formerly abandoned warehouses. In aggregate, these facilities house over 29,000 bitcoin miners with max capacity of over 900 PH/s (~0.7% of the network hashrate). The company on average pays approximately US$0.04 per kilowatt-hour (kWh) for its electricity, which is slightly above the estimated global industry median of $0.03/kWh according to research by BitOoda.

The company trades on the TSX Venture Exchange (TSXV) under the ticker BITF.

Bitfarms ($BITF) stock performance ($CAD)



coindesk *Source: Yahoo Finance, CoinDesk; Data as of July 24, 2020*

FINANCIALS

Balance sheet

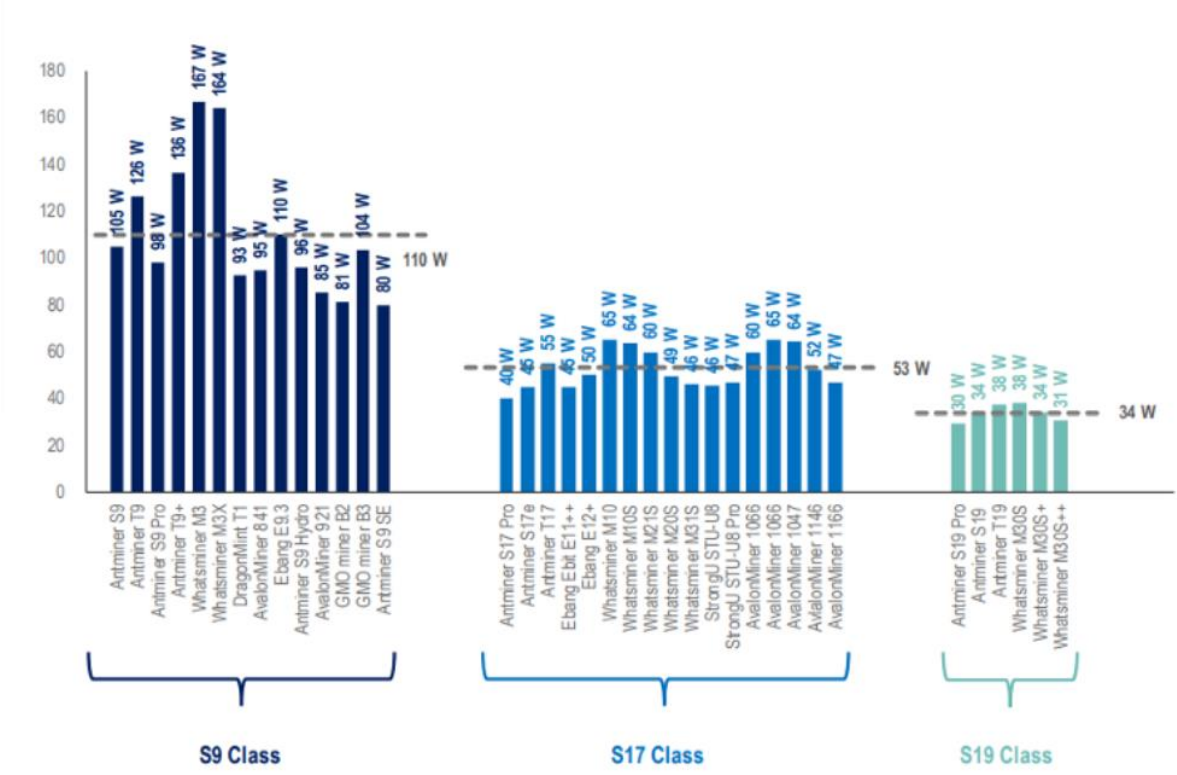
(YE 31-Dec, $USDm)	1Q19A	2Q19A	3Q19A	4Q19A	1Q20A
Assets					
Current assets					
Cash & cash equivalents	3.0	5.4	2.5	2.2	2.3
Trade receivables, net	0.5	0.9	1.0	1.1	0.7
Other assets	0.4	0.8	1.1	1.4	1.1
Inventories	0.4	0.3	0.2	0.1	0.3
Income taxes receivable	--	--	--	1.7	1.6
Assets held for sale	--	0.1	--	0.0	--
Total current assets	**4.3**	**7.4**	**4.7**	**6.6**	**6.1**
Non-current assets					
PP&E	21.4	24.2	35.1	37.8	36.0
Right-of-use assets	5.1	5.1	5.9	6.2	6.0
Intangible assets, net	0.8	0.7	0.6	0.7	0.6
Long-term deposits & advance payments on equipment	2.1	7.4	4.7	1.1	1.4
Embedded derivative	--	--	0.4	0.5	0.5
Total non-current assets	**29.4**	**37.4**	**46.8**	**46.3**	**44.6**
Total assets	**33.7**	**44.9**	**51.5**	**52.9**	**50.6**
Liabilities					
Current liabilities					
Current maturities of long-term debt	1.8	3.1	0.9	0.9	4.3
Current maturities of lease liabilities	1.0	0.7	0.6	0.7	0.7
Accounts payable and accrued liabilities	2.4	2.6	2.2	2.4	2.2
Income taxes payable	0.8	0.8	0.4	--	--
Total current liabilities	**6.0**	**7.1**	**4.1**	**4.0**	**7.2**
Non-current liabilities					
Long-term debt	3.7	9.6	14.7	14.8	11.4
Lease liabilities	4.0	4.5	5.3	5.7	5.1
Embedded derivative	--	0.4	--	--	--
Asset retirement provision	0.2	0.2	0.2	0.2	0.2
Warrant liability	0.9	4.2	--	--	--
Total non-current liabilities	**8.9**	**18.8**	**20.2**	**20.8**	**16.6**
Total liabilities	**14.9**	**26.0**	**24.3**	**24.8**	**23.8**
Shareholders equity					
Total shareholders equity	**14.2**	**15.6**	**27.2**	**28.1**	**26.9**
Non-controlling interest	**4.6**	**3.3**	**--**	**--**	**--**
Total liabilities & shareholders equity	**33.7**	**44.9**	**51.5**	**52.9**	**50.6**

Source: Bitfarms, FactSet, CoinDesk

Assets

By far, the largest component of Bitfarms' assets is its property, plant, and equipment (PP&E), the majority of which is comprised of its 29,000 ASIC miners.

Classification of bitcoin mining equipment (measured by W/TH)



Source: BitOoda

Of the 29,000 miners, Bitfarms owns approximately 15,000 of the quickly-aging Antminer S9's. The remainder mainly includes a mix of miners within the "S17 class" (a generation of miners more efficient than the S9's yet not as efficient as the newest models within the S19 class) purchased throughout 2019 and 2020.

Unlike some of its mining competitors, Bitfarms does not hold onto the bitcoin it mines. Instead, the company typically sells its production of bitcoin mined every few days as a conscious effort to reduce operational risk. At the end of 1Q20, Bitfarms only held 15 bitcoin on the balance sheet (worth approximately $144,000 based on current market prices).

Debt

Bitfarms' 2019 growth was financed primarily from a $20 million secured debt financing facility with New York-based Dominion Capital LLC. The facility was structured into four separate $5 million loan

tranches each with a 24-month maturity ending in a balloon payment for any remaining balance at the end of the term.

For each tranche utilized, Bitfarms also agreed to issue Dominion 1,666,667 warrants redeemable for Bitfarms equity shares at an exercise price of $0.40 USD per share.

Bitfarms drew down on all four $5 million loan tranches in March, April, June and August 2019.

Typically, the loans miners have been able to obtain include high interest rates and restrictive covenants. This one is no different with each tranche bearing 10% interest per annum. Unlike most loans to mining companies, however, this loan is not collateralized by bitcoin. Instead the loan is fully secured by the Bitfarms assets (which are mainly comprised of mining equipment) on a first-priority basis.

On the plus side, the structure of the loan allowed Bitfarms to fully deploy liquid capital into capital expenditures without having to lock up its bitcoin in collateral. The downside, however, came in the form of covenants & restrictions that included:

- Maintaining a bank balance of at least $1 million at all times
- Restrictions on issuing additional debt or selling property, plant, and equipment (PP&E) without re-investing the proceeds into new PP&E
- Restricted from issuing new common shares of equity below $0.40 USD per share

Income statement

(YE 31-Dec, $USDm)	1Q19A	2Q19A	3Q19A	4Q19A	1Q20A
Revenue	**3.6**	**8.5**	**9.7**	**10.5**	**9.2**
QoQ growth (%)	*-22%*	*135%*	*14%*	*8%*	*-13%*
# of BTC sold	*784*	*1,001*	*881*	*1,219*	*1,072*
QoQ growth (%)	*-1%*	*28%*	*-12%*	*38%*	*-12%*
Avg. price of BTC sold	*$4,629*	*$8,508*	*$11,054*	*$8,643*	*$8,593*
Bitfarms period-end hashrate at max capacity (PH/s)	*218*	*308*	*548*	*780*	*780*
QoQ growth (%)	*3%*	*41%*	*78%*	*42%*	*0%*
Network period-end hashrate (EH/s)	*46*	*62*	*89*	*95*	*115*
QoQ growth (%)	*13%*	*35%*	*44%*	*7%*	*21%*
Cost of revenue	***3.4***	***4.6***	***5.3***	***7.7***	***7.9***
Gross profit	**0.2**	**3.9**	**4.5**	**2.8**	**1.3**
G&A and other expenses	1.9	3.4	2.3	2.7	2.8
Impairment of PP&E & Intangibles	--	--	0.1	--	--
Operating income (loss)	**(1.7)**	**0.6**	**2.1**	**0.1**	**(1.5)**
Adjusted EBITDA	**(0.4)**	**3.3**	**4.7**	**3.0**	**2.8**
Adj. EBITDA margin (%)	*-12%*	*39%*	*48%*	*28%*	*30%*
Other comprehensive income	**(0.3)**	**(1.9)**	**2.2**	**(1.6)**	**(1.0)**
Pre-tax loss	**(2.0)**	**(1.3)**	**4.3**	**(1.4)**	**(2.5)**
Income tax expense (recovery)	--	--	(0.0)	(2.6)	(0.1)
Net income (loss)	**(2.0)**	**(1.3)**	**4.3**	**1.1**	**(2.4)**
Less: non-controlling interests	(0.7)	(0.5)	0.4	--	--
Net income attributable to equity shareholders	**(1.3)**	**(0.8)**	**3.9**	**1.1**	**(2.4)**
Net income (loss) per share					
Basic	$ (0.02)	$ (0.01)	$ 0.06	$ 0.02	$ (0.03)
Diluted	$ (0.02)	$ (0.01)	$ 0.05	NA	$ (0.03)
Weighted avg. shares outstanding					
Basic	*57.1*	*57.1*	*70.4*	*NA*	*84.0*
Diluted	*57.1*	*57.1*	*79.6*	*NA*	*90.1*

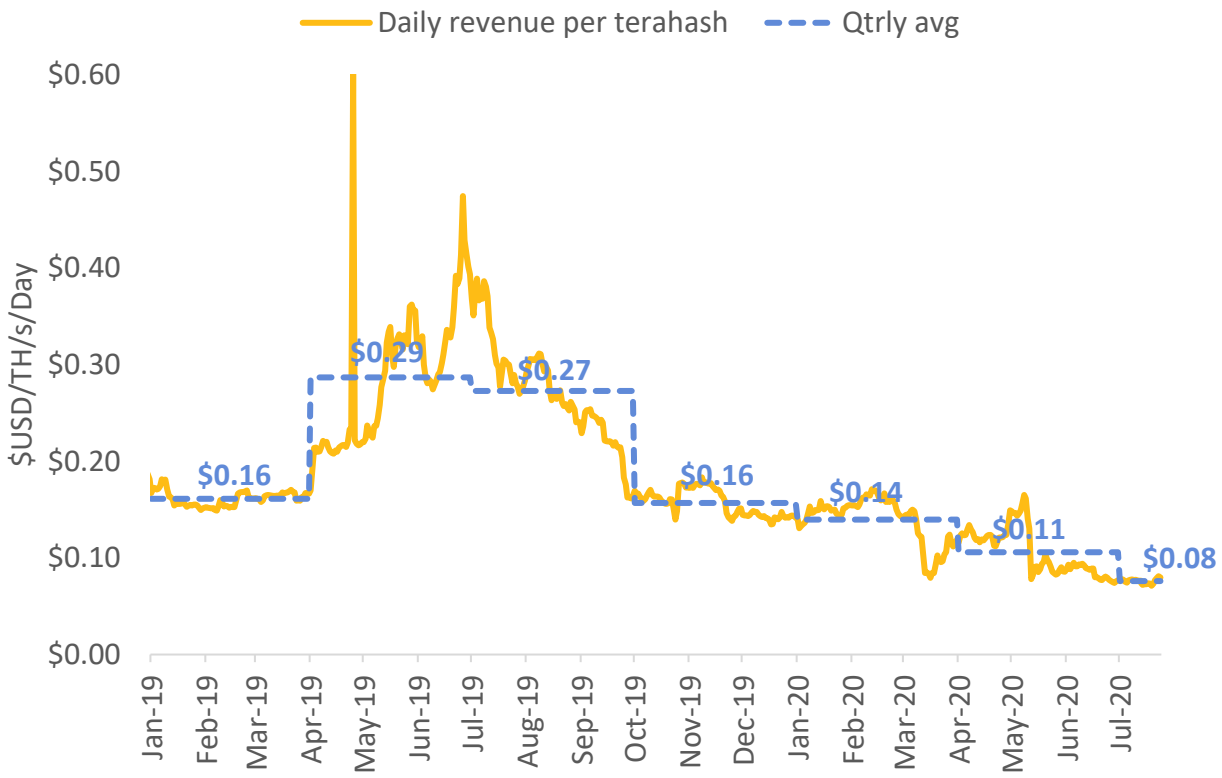
Source: Bitfarms, FactSet, CoinDesk

As Bitfarms continued its buildout of its ASIC miner portfolio throughout 2019, Bitfarms' revenue expanded from $3.6 million in 1Q19 to $10.5 million in 4Q19 as the company's hashrate growth exceeded that of the overall bitcoin network. Revenue growth was also bolstered by a 92% increase in bitcoin's price in 2019. Bitfarms' reported adjusted EBITDA margins improved from -10% to 28% within that time frame.

In 1Q20, revenue experienced a 13% sequential decrease caused by strong growth in the bitcoin network hashrate.

LOOKING AHEAD

Significant decline in revenue per TH since last year



coindesk *Source: HashrateIndex.com, CoinDesk*

Revenue per TH is a metric used to estimate the daily revenue ($USD) a bitcoin miner generates per terahash/second. Simply put, it measures the value of bitcoin's hashrate.

Since mid-2019, revenue per TH has declined considerably, causing strain for bitcoin miners across the industry. This reduction in revenue per TH was mainly attributed to the bitcoin network's increased difficulty rate as more efficient mining equipment came online coupled with the reduced block reward subsidy after the halving on May 11. As of July, daily revenue has averaged at $0.08 per TH, down 45% since 1Q20's daily average of $0.14 revenue per TH.

As a result, Bitfarms will likely experience a significant reduction in its adjusted EBITDA margin in 2Q20 and beyond based on the current scale of its operations assuming bitcoin prices hold steady.

MAJOR HEADWINDS

Debt coming due in 2021

Bitfarms' largest headwind is that the four $5 million tranches of its loan with Dominion Capital will all expire in 2021.

Upcoming financial obligations ($USDm)

Year	Long-term debt	Lease Liabilities	Total
2020	$2.7	$0.8	$3.5
2021	$17.9	$1.1	$19.0
2022 & beyond	$0.1	$5.3	$5.4

Source: Bitfarms, FactSet, CoinDesk

When adding together Bitfarms' future debt payments coupled with its future financial obligations on its lease liabilities, the company faces approximately $22.5 million in total financial obligations coming due by the end of 2021 according to its recent filings.

Estimated time needed for Bitfarms to pay off its upcoming financial obligations

Financial obligations due by end of 2021	$22.5
Less: Cash & Cash equivalents	$2.3
Less: Bitcoin in inventory	$0.1
Net debt	$20.1
1Q20 Adjusted EBITDA annualized	$11.1
Net debt / adjusted EBITDA	**1.8 years**

Source: Bitfarms, FactSet, CoinDesk

The net debt-to-adjusted EBITDA ratio is a common ratio used to estimate how long it would take a company to pay off its financial obligations by comparing the company's net debt to its cash earnings.

In 1Q20 (Bitfarms' last reported financials), the company earned $2.8 million in adjusted EBITDA. Assuming Bitfarms was able to maintain this level of adjusted EBITDA until the end of 2021, the company would take approximately 1.8 years to amass enough capital to pay off its debt. In other words, the company would struggle to make due on its payments at current output of cash earnings.

As stated previously, however, we estimate that Bitfarms' adjusted EBITDA will be reduced significantly moving forward as a result of a reduction in the bitcoin mining industry's revenue per TH, making it

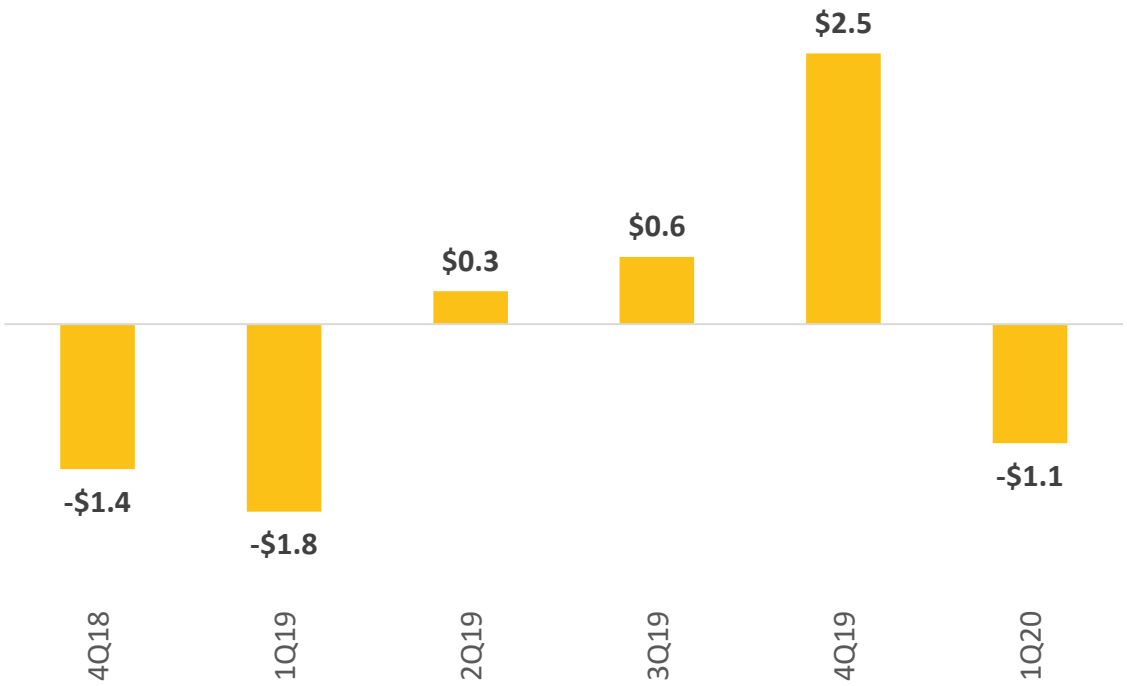
considerably more difficult for Bitfarms to meet its financial obligations based on the size of its current operations.

Company will likely need to expand operations, but lacks the capital to do so

As a result of the downward pressure on Bitfarms' mining profitability, the company will likely need to expand its operations with additional "newer" generation equipment in order to be able to pay off its eminent balloon payments on debt coming due in 2021.

However, with only $2.3 million in cash as of 1Q20, Bitfarms currently lacks the capital needed to expand operations in any meaningful capacity.

Bitfarms' quarterly net working capital ($USDm)



coindesk *Source: Bitfarms, Factset, CoinDesk*

Adding to the issue, noise complaints from local residents near Bitfarms' largest facility have proven to be a headache with the City of Sherbrooke threatening legal action if proper measures are not taken to reduce the sound emission. The company has since made progress reducing the noise as it completed the construction of a sound barrier wall which cost approximately $0.2 million according to recent filings. However, there is still more construction work to be done to fully satisfy city officials.

In conversations with CoinDesk, Bitfarms' management believes that the total estimated cost of noise reduction will ultimately be immaterial to the business. However, at the very least, this will serve as a short-term capital drain for a company in need of expanding its mining operations.

Limited options for raising capital

Due to the restrictions and covenants attached to Bitfarms' debt facility, the company may have limited existing options for raising additional capital.

As per the agreement with Dominion Capital, Bitfarms cannot issue new shares of equity below US$0.40 a share for the entire term of the loan. With shares currently trading at C$0.355 (US$0.265), raising capital through issuing new shares of common stock seems to be off the table.

Raising capital via debt will also be extremely difficult with all of Bitfarms' assets currently locked up in collateral with Dominion.

On the 1Q20 earnings call from last month, management noted alternative methods of raising funds such as equipment financing (i.e. loans collateralized by newly purchased ASIC miners) as a possibility. However, this form of lending has not been that prevalent within the mining industry as it's much easier for lenders to seize and sell off bitcoin collateral compared to ASIC miners.

Even if Bitfarms were able to obtain such a deal, the loan would likely come with high interest rates, adding to the company's existing debt obligations.

CONCLUSION

Canadian-based Bitfarms finds itself in a race against time. At its core, it operates decent equipment at a respectable cost of electricity resulting in positive operating cash flows. However, the company used high-interest rate debt with large balloon payments in order to grow its operations. Now with over $20 million in financial obligations coming due by the end of 2021 coupled with declining revenue output per terahash, Bitfarms will likely struggle to pay off its debt.

Assuming there's no significant jump in bitcoin prices, Bitfarms will probably need to expand operations with efficient mining equipment within the next twelve months which will require the company to raise additional capital. However, due to the covenants and restrictions from its existing loan, the company's options for raising funds may be limited as the company can't issue new common equity at its current market price and lacks the non-collateralized assets generally needed to obtain new loans.

UPDATE (July 28, 03:00 UTC): The wording around Bitfarms' ability to raise capital was modified to make clear it expresses the analyst's opinion.